It follows that the function $L_j : \mathbb{R}^{2\times 2}_{\rm sym} \to \mathbb{R}^3$ that maps a matrix $D \in \mathbb{R}^{2\times 2}_{\rm sym}$ into the vector $L_j(D)$ such that

$$\bar{Q}_j(D) = Q_j(U_j^{-1}(D + L_j(D) \otimes e_3 + e_3 \otimes L_j(D))) \tag{9}$$

is well-defined owing to the coercivity of Q_j. Moreover, L_j is linear. Indeed, $Q_j(U_j^{-1}(\cdot))$ can be represented by a fourth-order tensor $\mathbb{C}$ so that

$$Q_j(U_j^{-1}(D + x \otimes e_3 + e_3 \otimes x)) = \mathbb{C}(D + x \otimes e_3 + e_3 \otimes x) : (D + x \otimes e_3 + e_3 \otimes x)\,.$$

In particular, $L_j(D)$ is the unique vector that solves the linear optimality conditions

$$\mathbb{C}(D + x \otimes e_3 + e_3 \otimes x) : (e_i \otimes e_3 + e_3 \otimes e_i) = 0, \qquad i = 1, 2, 3.$$

3 Compactness estimates

In order to study the Γ-convergence of the functionals E_h^α we first need to establish compactness for sequences of deformations that have bounded rescaled energy. It is clear that the elastic part of the rescaled energy forces the rescaled deformation gradient to approach in the limit the union of the wells K. However, we would like to prove that the rescaled gradients are actually getting closer to a single well K_i. This is precisely ensured by the penalty term. In the following result, inspired by [1], we give a precise meaning to this statement.

Proposition 3.1. *Let $\alpha \geq 2$. Let $(y_h) \subset W^{1,2}(\Omega; \mathbb{R}^3)$ be a sequence such that*

$$\lim_{h\to 0} h^\alpha E_h^\alpha(y_h) = 0\,. \tag{10}$$

Then for $h \ll 1$ there are an index $i_h \in \{1, \dots, l\}$ and two constants $\delta, C(\delta) > 0$ such that

$$\int_{\Omega_h^{i_h}} \mathrm{dist}^2(\nabla_h y_h, K_{i_h})\, dx \leq C(\delta) h^\alpha E_h^\alpha(y_h)\,, \tag{11}$$

$$\int_{\Omega\setminus\Omega_h^{i_h}} \mathrm{dist}^2(\nabla_h y_h, K_{i_h})\, dx \leq C(\delta) h^\alpha [(E_h^\alpha(y_h))^\theta + E_h^\alpha(y_h)]\,, \tag{12}$$

where $\Omega_h^{i_h} = \{x \in \Omega \,:\, \mathrm{dist}(\nabla_h y_h(x), K_{i_h}) \leq \delta\}$ and

$$\theta = \begin{cases} \frac{3}{2} & \text{if } q = 2\,, \\ \frac{5}{3} & \text{otherwise}\,. \end{cases}$$

The proof follows the same arguments used in [1, Theorem 2.3]. For the convenience of the reader, we give a sketch of the proof.

Proof. Fix $\delta < 1 \wedge \min_{i\neq j}\{\frac{1}{2}\,\mathrm{dist}(K_i, K_j)\}$ and define

$$\Omega_h = \{x \in \Omega\colon\ \mathrm{dist}(\nabla_h y_h(x), K) \leq \delta\} = \bigcup_{i=1}^{l} \Omega_h^i\,,$$

where

$$\Omega_h^i = \{x \in \Omega\colon\ \mathrm{dist}(\nabla_h y_h(x), K_i) \leq \delta\}\,.$$

Note that, by (W4), condition (11) holds with an arbitrary $j \in \{1, \dots, l\}$ in place of i_h.

Define $\widetilde{W}(A) = f_q(\mathrm{dist}(A,K))$ for $A \in \mathbb{R}^{3\times 3}$, where we recall that $f_q(t) = t^2 \wedge t^q$. Let

$$d_{\widetilde{W}}(A,B) = \inf\left\{\int_0^1 (\widetilde{W}(\xi(s)))^{\frac{l}{mp}} |\xi'(s)|\, ds \colon \xi \in C^1([0,1];\mathbb{R}^{d\times d}) \text{ s.t. } \xi(0) = A, \xi(1) = B\right\},$$

where $l, m > 1$ are such that $\frac{1}{l} + \frac{1}{m} = 1$. By (10) and (W4) we have that

$$|\Omega \backslash \Omega_h| \leq h^\alpha E_h(y_h) \to 0. \tag{13}$$

Hence, there exists $i_h \in \{1,\dots,l\}$ such that $|\Omega_h^{i_h}| \geq C(\delta)$. We define

$$g_h(x) = (d_{\widetilde{W}}(\nabla_h y_h(x), K_{i_h}) - \tilde{\delta}) \vee 0\,,$$

where $\tilde{\delta} = \delta^{2\frac{l}{mp}+1}$. One can check that $g_h \equiv 0$ on $\Omega_h^{i_h}$. Set $\beta = \frac{p}{l}$ and choose l in such a way that $\beta < 3$. Observe that $g_h \in W^{1,\beta}(\Omega)$ and

$$|\nabla g_h| \leq (\widetilde{W}(\nabla_h y_h))^{\frac{l}{mp}} |\nabla(\nabla_h y_h)|\,.$$

Hence, applying the Sobolev embedding, the Poincaré inequality given in [1, Lemma 2.2], and the Young inequality with exponents $\frac{1}{l}$ and $\frac{1}{m}$, we deduce that

$$\begin{aligned}\int_\Omega |g_h|^{\beta^*}\, dx &\leq C\left(\int_\Omega |\nabla g_h|^\beta\, dx\right)^{\frac{\beta^*}{\beta}} \leq \left(\int_\Omega (\widetilde{W}(\nabla_h y_h))^{\frac{1}{m}} |\nabla(\nabla_h y_h)|^\beta\, dx\right)^{\frac{\beta^*}{\beta}} \\ &\leq C\frac{h^{\alpha\frac{\beta^*}{\beta}}}{\eta(h)^{\beta^*}} E_h^\alpha(y_h)^{\frac{\beta^*}{\beta}}\,,\end{aligned} \tag{14}$$

where $\beta^* = \frac{3\beta}{3-\beta}$ is the critical Sobolev exponent in dimension 3. Note that here we used the crucial information that $|\Omega_h^{i_h}| \geq C(\delta)$ to deduce that the constant in the Poincaré inequality is independent of h. Let

$$\tilde{\Omega}_h^i = \{x \in \Omega \colon d_{\widetilde{W}}(\nabla_h y_h, K_i) \leq 2\tilde{\delta}\}\,.$$

By [1, Lemma 2.6], we can refine the choice of δ in such a way that $\tilde{\Omega}_h^{i_h} \backslash \Omega_h^{i_h} \subset \Omega \backslash \Omega_h \subset \Omega \backslash \Omega_h^{i_h}$. Moreover, by [1, Lemmas 2.5 and 2.6] we have that $\mathrm{dist}(\nabla_h y_h, K_{i_h}) \leq C g_h$ on $\Omega \backslash \tilde{\Omega}_h^{i_h}$ and $\mathrm{dist}(\nabla_h y_h, K_{i_h}) \leq C$ on $\tilde{\Omega}_h^{i_h} \backslash \Omega_h^{i_h}$. Thus, writing $\Omega \backslash \Omega_h^{i_h}$ as $(\Omega \backslash \tilde{\Omega}_h^{i_h}) \cup (\tilde{\Omega}_h^{i_h} \backslash \Omega_h^{i_h})$ we deduce from (13)–(14) that

$$\begin{aligned}\int_{\Omega\backslash\Omega_h^{i_h}} \mathrm{dist}^{\beta^*}(\nabla_h y_h, K_{i_h})\, dx &\leq C\int_{\Omega\backslash\tilde{\Omega}_h^{i_h}} |g_h|^{\beta^*}\, dx + C|\tilde{\Omega}_h^{i_h} \setminus \Omega_h^{i_h}| \\ &\leq C\left(\frac{h^{\alpha\frac{\beta^*}{\beta}}}{\eta(h)^{\beta^*}} E_h^\alpha(y_h)^{\frac{\beta^*}{\beta}} + h^\alpha E_h^\alpha(y_h)\right)\,.\end{aligned} \tag{15}$$

If $q \neq 2$, we choose $l = \frac{5}{6}p$. Note that $l > 1$ by (P3). Then $\beta = \frac{6}{5}$, $\beta^* = 2$, and $\frac{\beta^*}{\beta} = \frac{5}{3} = \theta$. By (P1) and (15) we get (12).

If $q = 2$, we choose $l = p$, so that $\beta = 1$, $\beta^* = \frac{3}{2}$ and $\frac{\beta^*}{\beta} = \theta = \frac{3}{2}$. Fix a constant $M > 0$ such that $K \subset B_M(0)$ and define $B_h^M = \{x \in \Omega \colon |\nabla_h y_h(x)| \leq M\}$. Writing $\Omega \backslash \Omega_h^{i_h}$ as $((\Omega \backslash \Omega_h^{i_h}) \cap B_h^M) \cup ((\Omega \backslash \Omega_h^{i_h}) \backslash B_h^M)$, by (W4) we deduce that

$$\begin{aligned}\int_{\Omega\backslash\Omega_h^{i_h}} \mathrm{dist}^2(\nabla_h y_h, K_{i_h})\, dx &\leq C\int_{(\Omega\backslash\Omega_h^{i_h})\cap B_h^M} \mathrm{dist}^{\frac{3}{2}}(\nabla_h y_h, K_{i_h})\, dx \\ &\quad + C\int_{(\Omega\backslash\Omega_h^{i_h})\backslash B_h^M} W(\nabla_h y_h)\, dx\,.\end{aligned} \tag{16}$$

Then, (12) follows by (P1) and (15)–(16). ■

Remark 3.2. At a first glance, it might seem that the hypothesis (10) of Proposition 3.1 does not depend on α, while the thesis (12) does. However, (P1) prescribes a dependence on α of the penalty term coefficient $\eta(h)$. This is particularly clear examining the sketch of the proof above.

Corollary 3.3. *Let $\alpha \geq 2$. Let $(y_h) \subset W^{1,2}(\Omega;\mathbb{R}^3)$ be a sequence such that*

$$\lim_{h\to 0} h^\alpha E_h^\alpha(y_h) = 0\,.$$

Then for $h \ll 1$ there are an index $i_h \in \{1,\dots,l\}$ and two constants $\delta, C(\delta) > 0$ such that

$$\int_\Omega \operatorname{dist}^2(\nabla_h y_h, K_{i_h})\,dx \leq C(\delta) h^\alpha E_h^\alpha(y_h).$$

The following is a variant of the well-known rigidity estimate by Friesecke, James, and Müller (see [16, 15]), where the well $\mathrm{SO}(3)$ is replaced by $K_j = \mathrm{SO}(3)U_j$.

Proposition 3.4. *Let $(y_h) \subset W^{1,2}(\Omega;\mathbb{R}^3)$ and let $j \in \{1,\dots,l\}$. Define*

$$D_{h,j} = \|\operatorname{dist}(\nabla_h y_h, K_j)\|_{L^2(\Omega)}\,.$$

There are two maps $R_h \in L^\infty(S;\mathrm{SO}(3))$ and $\tilde{R}_h \in W^{1,2}(S;\mathbb{R}^{3\times 3}) \cap L^\infty(S;\mathbb{R}^{3\times 3})$ such that

(R1) $\|\nabla_h y_h - R_h U_j\|_{L^2(\Omega)} \leq C D_{h,j}$,

(R2) $\|\nabla \tilde{R}_h\|_{L^2(S)} \leq C h^{-1} D_{h,j}$,

(R3) $\|\tilde{R}_h - R_h\|_{L^2(S)} \leq C D_{h,j}$,

(R4) $\|\tilde{R}_h - R_h\|_{L^\infty(S)} \leq C h^{-1} D_{h,j}$.

Moreover, there exists a constant rotation $Q_h \in \mathrm{SO}(3)$ such that

$$\|R_h - Q_h\|_{L^2(S)} \leq C h^{-1} D_{h,j}\,.$$

Finally, if $h^{-1} D_{h,j} \to 0$, then for $h \ll 1$ we can choose $\tilde{R}_h = R_h$.

To prove this result it is enough to follow the same approach of [15]. Indeed, the rigidity estimate [15, Theorem 5] holds also for a well of the form $\mathrm{SO}(3)U_j$ by a change of variable. Then, all the estimates in [15, Theorem 6] can be carried out in the same fashion.

Remark 3.5. If $r > 1$, all the results of Proposition 3.4 hold with the L^2 norm replaced by the L^r norm and the factor h^{-1} replaced by $h^{-\frac{2}{r}}$.

3.1 Compactness in the Kirchhoff case $\alpha = 2$

In the case $\alpha = 2$ the Γ-limit is written in terms of the deformation gradient. In this section we show compactness for sequences of rescaled gradients and give a characterization of their limit. Firstly, we will need an explicit expression of the vector ν that appears in (3).